

Servo Saver - CRC Cards

FuelStation

Class: FuelStation - Front	
Superclass: none Sub-class: none	
Responsibilities	Collaborators
Know station identity, brand and opening hours	-
Provide current prices for search results	FuelPrice, FuelType, SearchRequest
Maintain price history and locked daily price caps	FuelPrice, DailyPriceCap
Provide station details and position for map / directions	Location
Receive and hold reports raised about the station	Report (IssueReport, FuelUnavailabilityReport)
Know its owning retailer	FuelRetailer

Class: FuelStation - Back
Attributes - stationId : String - name : String - brand : String - openingHours : String - address : String - isOperating : Boolean - isMembersOnly : Boolean
Relationships Generalisation: none Association: - Location (1 to 1); - SearchRequest (0..* to 0..*); - Report (1 to 0..*) Aggregation: part of FuelRetailer (owner 1 to 1..* stations) Composition: whole of FuelPrice (1 to 0..*); whole of DailyPriceCap (1 to 0..*)

Business Rules

- A station can only be registered by a FuelRetailer that has been approved by ServiceVictoriaAdmin.
- A station holds at most one active FuelPrice per FuelType at any time.
- A station appears in search results only when it lies within the search radius of the request.

FuelPrice

Class: FuelPrice - Front	
Superclass: none Sub-class: none	
Responsibilities	Collaborators
Know its amount, submission time and lifecycle status	-
Validate the submitted amount against the locked daily cap	DailyPriceCap
Activate a validated price so it is shown to motorists	FuelStation
Expire / supersede itself when a newer price is activated or it goes stale	FuelStation, Scheduler (flag stale)
Know which fuel type it prices	FuelType
Supply price data to search results	SearchRequest

Class: FuelPrice - Back
Attributes - priceId : String - amount : Float - submittedAt : DateTime - status : String (submitted / validating / active / rejected / stale)
Relationships Generalisation: none Association: - FuelType (0..* prices to 1 fuelType); - DailyPriceCap (0..* checkedPrices to 1 appliedCap) Aggregation: none Composition: part of FuelStation (station 1 to 0..* priceHistory)
Business Rules - A submitted amount must not exceed the locked DailyPriceCap for that fuel type and effective date. - Only a validated price can become active; activating a new price supersedes the previous active price for that fuel type. - A price not refreshed by the retailer is flagged stale by the Scheduler and removed from display.